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# OSF project metadata — The Coupled Co-Scaling Law

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## Title

The Coupled Co-Scaling Law: A Falsifiable Threshold Criterion for the Stability of Recursive Self-Improvement

## Contributor

Michael Darius Eastwood (independent researcher; ARC/Eden research programme). Add your ORCID if you have one.

## Category

Project

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## Description

This project archives the working paper "The Coupled Co-Scaling Law" together with its complete computational apparatus: a runnable verification harness, independent theorem test suites, a pre-registered real-model experiment protocol with its harness, and a SHA-256 integrity manifest.

Most work on recursive self-improvement treats it as a problem of speed: capability may grow explosively and outrun supervision, so the proposed lever is to cap the growth rate. From a minimal dynamical model, this paper derives that the operative lever is different. Stability is set not by the growth rate but by a single inequality between two scaling exponents: the rate at which a misalignment-correcting force strengthens with capability ( $\beta$ ) must exceed the rate at which drift accelerates with capability ( $k$ ). The criterion is  $\beta > k$ .

Its sharpest consequence addresses the central fear of the field: a hard takeoff, even a genuine finite-time intelligence explosion, drives the modelled misalignment fraction to zero if and only if  $\beta > k$ , and the speed of the explosion does not change that asymptotic verdict. The criterion shares the threshold form of the quantum error-correction sub-threshold condition, offered as a falsifiable hypothesis (the model's suppression law is power-law, not exponential). The closed-form predictions are checked by a runnable harness, and the theorem statements are independently re-derived in a standalone test suite.

Scope and honesty. The results hold within the stated minimal model. The decisive empirical question, whether real self-improving systems satisfy the criterion, remains the open problem; a pre-registered protocol and a runnable real-model harness are included to make that test possible. The paper has been hardened against three independent adversarial reviews.

Author: Michael Darius Eastwood (independent researcher; ARC/Eden research programme). Working paper dated 26 June 2026; deposited to OSF 29 June 2026. Code and full commit history:

<https://github.com/MichaelDariusEastwood/arc-principle-validation>

## Tags

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AI safety; AI alignment; recursive self-improvement; intelligence explosion; hard takeoff; scalable oversight; scaling laws; dynamical systems; Lyapunov stability; threshold criterion; quantum error correction analogy; falsifiability; AI governance; control theory; co-scaling

## Subjects (OSF / bepress taxonomy)

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- Physical Sciences and Mathematics > Computer Sciences > Artificial Intelligence and Robotics
- Physical Sciences and Mathematics > Computer Sciences > Theory and Algorithms
- Physical Sciences and Mathematics > Applied Mathematics > Dynamic Systems

## Citation

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Eastwood, M. D. (2026). The Coupled Co-Scaling Law: A Falsifiable Threshold Criterion for the Stability of Recursive Self-Improvement. ARC/Eden research programme. OSF.  
<https://doi.org/10.17605/OSF.IO/6C5XB>

## Dates

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- Document (priority) date: 26 June 2026 — leave untouched on the paper.
- OSF deposit date: 29 June 2026 — recorded automatically by OSF; do not backdate.